

Ayush Patel

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LinkedIn | GitHub | Portfolio

Unjha, Gujarat, India

Summary

Software Engineer and GenAI Engineer Intern skilled in full-stack development (Python, React.js, Node.js) and LLM-powered multi-agent systems (LangChain, LangGraph, RAG). 220+ DSA problems solved.

Education

Adani University, Ahmedabad, Gujarat

June 2023 – Present

Bachelor of Technology in Computer Science & Engineering (AI & ML)

CGPA: 8.72 / 10.00

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Deep Learning, Cyber Security, Agile Development, DevOps.

Technical Skills

- **Languages:** C, Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Tailwind CSS, FastAPI, Socket.IO, WebRTC
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **AI / GenAI:** PyTorch, OpenCV, LangChain, LangGraph, LangSmith, MCP, RAG, LLMs, Vector Databases, Agentic AI, Prompt Engineering, Transformer Models
- **Developer Tools:** Git, GitHub, Docker, Linux, VS Code, Postman
- **Core Concepts:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, System Design, Problem Solving

Experience

GenAI Engineer Intern

May 2026 – Present

OxiqAI — Remote, Pune, India

- Developed production-ready LLM-powered applications using LangChain, LangGraph, and MCP, applying agentic AI architecture to automate enterprise workflows.
- Designed modular AI agents and workflow pipelines using object-oriented principles, improving code reusability across the platform.
- Built multi-agent RAG systems with tool calling, memory, and workflow orchestration, cutting manual information-retrieval time by roughly 40%.
- Engineered scalable FastAPI backend services with PostgreSQL-backed REST APIs, and implemented observability, fallback strategies, and guardrails to strengthen production reliability.
- Automated a fully autonomous HRMS platform – covering leave management, payroll, attendance, and employee records – using asyncio-based concurrent AI agents, cutting manual HR processing time by roughly 50%.

Projects

SynAgent – Multi-Agent Research System

LangChain, LangGraph, Tavily API,

BeautifulSoup, Python, Streamlit

Live Demo | GitHub

- Designed a multi-agent research platform that autonomously searches, extracts, and summarizes web content using LangChain and LangGraph-based agent workflows.
- Integrated Tavily Search API and BeautifulSoup-based scraping to pull live data across 10+ web sources per query.
- Built an interactive Streamlit interface delivering low-latency, structured AI research reports.

QuolaGPT – Multilingual Transformer Language Model (Built From Scratch)

PyTorch, FastAPI,

React.js, NLP

Live Demo | GitHub

- Built a GPT-style Transformer language model from scratch in PyTorch, implementing a custom tokenizer, multi-head self-attention, positional encoding, and decoder blocks – without HuggingFace Transformers or OpenAI APIs.
- Developed complete training and inference pipelines covering preprocessing, batching, checkpointing, and evaluation.
- Trained multilingual models on English, Hindi, and Gujarati corpora, and deployed inference via a FastAPI backend with a React.js frontend.

Real-Time Location Sharing Web Application

Node.js, Express.js, Socket.IO,

WebRTC, Leaflet.js

Live Demo | GitHub

- Architected a live location-sharing platform using Node.js, Express.js, Socket.IO, and WebRTC, supporting 50+ concurrent users.
- Implemented bidirectional WebSocket sync for continuous location updates, visualized through Leaflet.js with dynamic markers.
- Designed a room-based architecture and deployed the frontend on Vercel with the backend on Render.

Achievements

- Solved **220+ Data Structures & Algorithms** problems across coding platforms, strengthening problem-solving and algorithmic thinking.
- Secured **2nd place** in the CodeCrypts programming competition (C language), demonstrating strong analytical and debugging skills.
- Participated in Smart India Hackathon (SIH) 2024, HackOut 2025 (DA-IICT), and Odoo Hackathon × Adani University 2026.